

1.1 Quadratics

Pure Mathematics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Completing the square reveals turning point and range; discriminant b^2-4ac classifies real roots. A quadratic inequality changes sign at roots when they exist; use a sketch or interval sign test, including endpoints as the question requires.

Worked example: follow the reasoning

$x^2-6x+5=(x-3)^2-4$, so minimum -4 at $x=3$ and roots $1,5$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Solve $x^2-5x+6<0$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Factor $(x-2)(x-3)<0$ [1]; upward parabola negative between roots [1]; $2<x<3$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.